THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a pie chart of 216 students a sector has a central angle of 100°. The number of students it represents is: 1

(a) 63 **(b)** 120 **(c)** 10 **(d)** 60

2. If A:B = 4:9 and B:C = 3:10, then A:C is: 1

(a) 4:31 **(b)** 2:15 **(c)** 4:10 **(d)** 40:27

3. 8 workers working 9 hours a day finish a job in 25 days. 15 workers working 6 hours a day will finish it in (days): 1

(a) 20 **(b)** 13.33 **(c)** 22 **(d)** 200

4. The ratio of boys to girls in a school is 4:7. If there are 990 students, the number of girls is: 1

(a) 360 **(b)** 495 **(c)** 979 **(d)** 630

5. A sum of ₹ 2100 buys 28 kg of rice at ₹ 75 per kg. When the price falls to ₹ 60 per kg, the same money buys (kg): 1

(a) 33 **(b)** 35 **(c)** 28 **(d)** 56

6. A train at 48 km/h takes 15 hours for a journey. At 60 km/h it will take (hours): 1

(a) 18.75 **(b)** 11 **(c)** 13 **(d)** 12

7. Two numbers are in the ratio 4:13 and their sum is 170. Their difference is: 1

(a) 92 **(b)** 17 **(c)** 85 **(d)** 90

8. In a pie chart of 126 students a sector has a central angle of 60°. The number of students it represents is: 1

(a) 21 **(b)** 24 **(c)** 6 **(d)** 42

9. If A:B = 7:8 and B:C = 12:5, then A:C is: 1

(a) 35:96 **(b)** 7:5 **(c)** 21:10 **(d)** 21:11

10. 14 workers working 8 hours a day finish a job in 21 days. 7 workers working 12 hours a day will finish it in (days): 1

(a) 294 **(b)** 30 **(c)** 28 **(d)** 42

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A pie chart of 160 students has a sector of 135°. How many students does it represent, and what percent of the students is that? 2

12. Two numbers are in the ratio 6:13 and their sum is 190. Find the numbers. 2

13. A hostel has food for 72 students for 15 days. How long will the food last if 12 students leave? 2

14. A pie chart of 96 students has a sector of 150°. How many students does it represent, and what percent of the students is that? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 9 workers working 10 hours a day can finish a job in 16 days. In how many days can 12 workers working 6 hours a day finish it? 3

16. A:B = 1:3 and B:C = 2:3. If A+B+C = 170, find the value of each. 3

17. In a pie chart, the sector for Dance measures 30° and represents 9 students. Find the total number of students and the number of students represented by a 150° sector. 3

OR

18 workers working 6 hours a day can finish a job in 25 days. In how many days can 15 workers working 9 hours a day finish it? 3

18. A:B = 5:6 and B:C = 3:4. If A+B+C = 190, find the value of each. 3

OR

In a pie chart, the sector for Chess measures 36° and represents 7 students. Find the total number of students and the number of students represented by a 72° sector. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A mixture of 72 litres has milk and water in the ratio 1:5. How much water must be added to make the ratio 1:8? 5

OR

8 workers can complete a task in 36 days. After 12 days, 4 more workers join them. In how many more days will the work be finished? 5

20. A mixture of 75 litres has milk and water in the ratio 2:3. How much water must be added to make the ratio 2:5? 5

OR

18 workers can complete a task in 25 days. After 10 days, 12 more workers join them. In how many more days will the work be finished? 5

END OF QUESTION PAPER — Total: 40 Marks

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